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(19) **United States**(12) **Patent Application Publication**
DUAN et al.(10) **Pub. No.: US 2025/0281444 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **COMPOUND WITH NEW SKELETON OF URSANE TRITERPENOID - ENT-KAURANE DITERPENOID DIMER, AND PREPARATION METHOD THEREOF AND APPLICATION THEREOF**(71) Applicant: **NANJING UNIVERSITY OF CHINESE MEDICINE**, Nanjing (CN)(72) Inventors: **Jinao DUAN**, Nanjing (CN); **Ming ZHAO**, Nanjing (CN); **Hongkai BI**, Nanjing (CN); **Junfei ZHOU**, Nanjing (CN); **Cai LU**, Nanjing (CN); **Chuanlin JIN**, Nanjing (CN); **Mengyu DAI**, Nanjing (CN); **Sheng GUO**, Nanjing (CN); **Shulan SU**, Nanjing (CN); **Hui YAN**, Nanjing (CN)(73) Assignee: **NANJING UNIVERSITY OF CHINESE MEDICINE**, Nanjing (CN)(21) Appl. No.: **19/215,404**(22) Filed: **May 22, 2025**(30) **Foreign Application Priority Data**

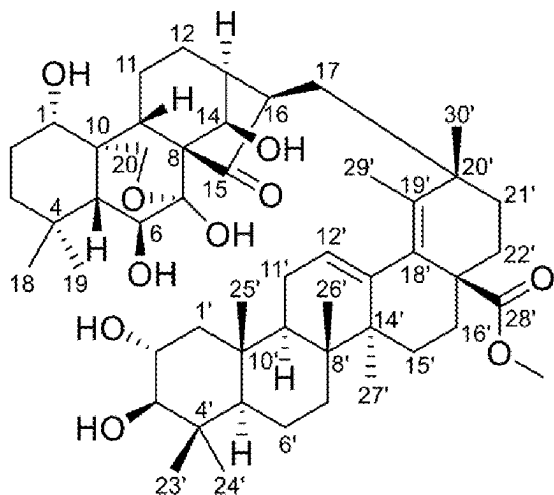
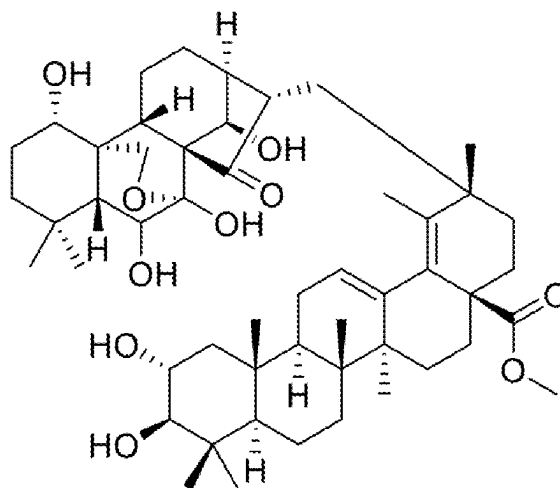
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(57)

ABSTRACT

Disclosed are two compounds with a new skeleton of ursane triterpenoid—ent-kaurane diterpenoid dimer and a preparation method thereof. According to the invention, chemical ingredients of a tuber of a West African plant *I. trichantha* are deeply studied, and separated to obtain two compounds Bisicacicol B (1) and Bisicacicol C (2), and the compounds are both a natural hybrid dimer formed by a DielsAlder addition or Michael addition reaction between a ursane triterpenoid with a unique spiro structure and an ent-kaurane diterpenoid. Pharmaceutical experiments show that the Bisicacicol B (1) and Bisicacicol C (2) prepared by the invention have a significant anti-tumor activity (on a pancreatic cancer, a colon cancer, a lung cancer, or a liver cancer), and can also be combined with an amphotericin B to exert a coordinated and synergistic anti-*Candida albicans* activity.

**1****2**

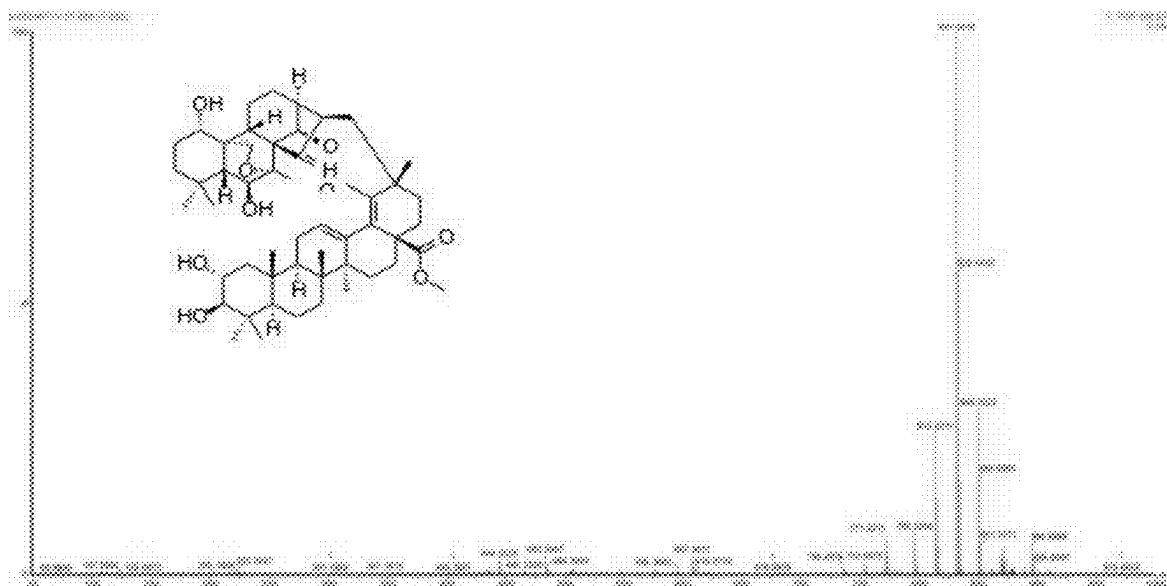


FIG. 1

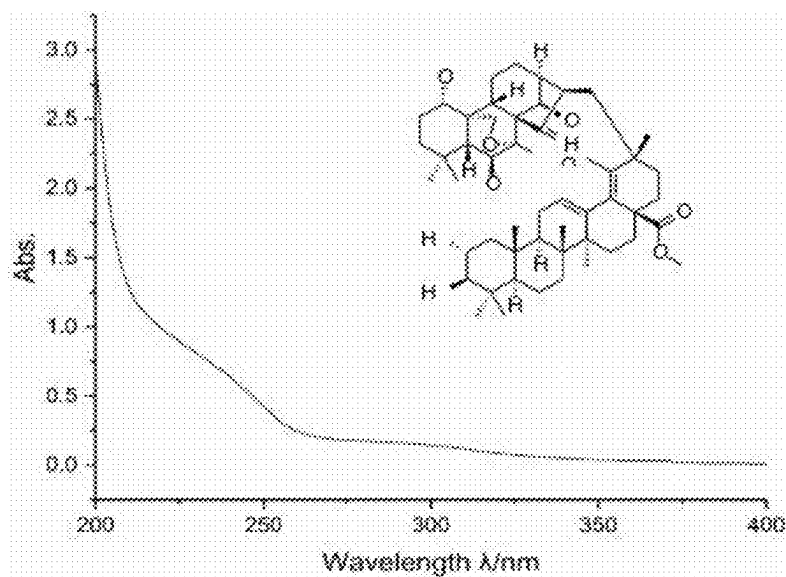


FIG. 2

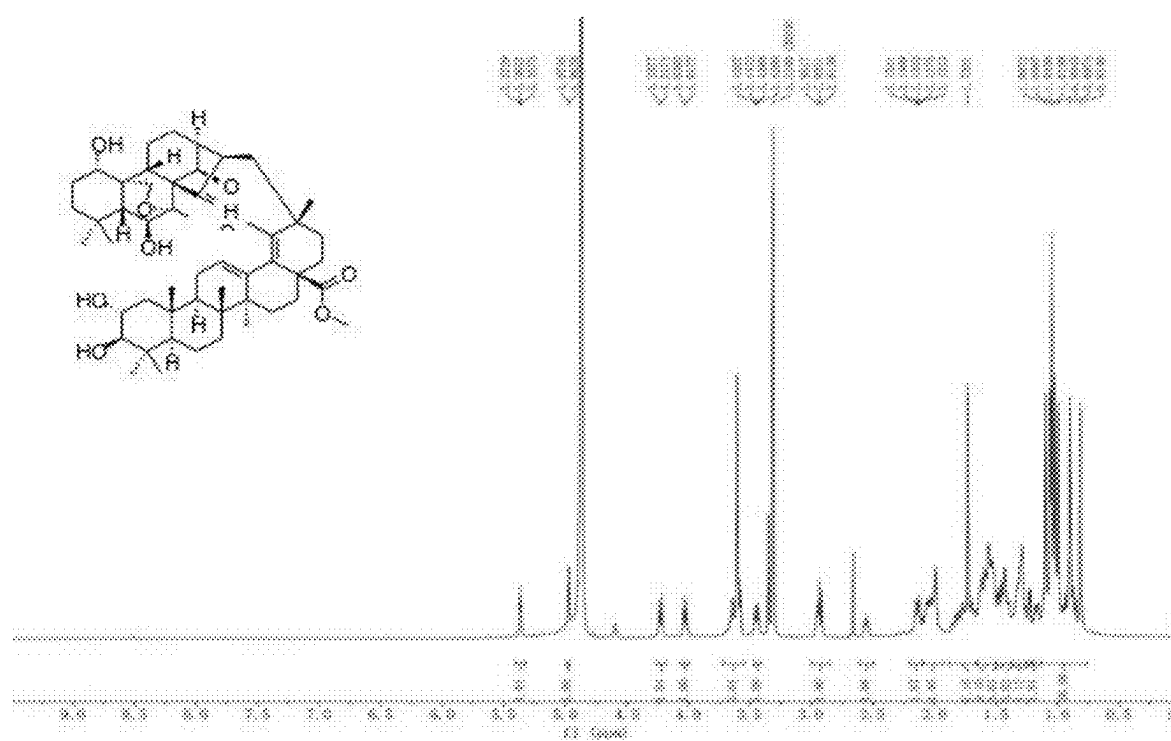


FIG. 3

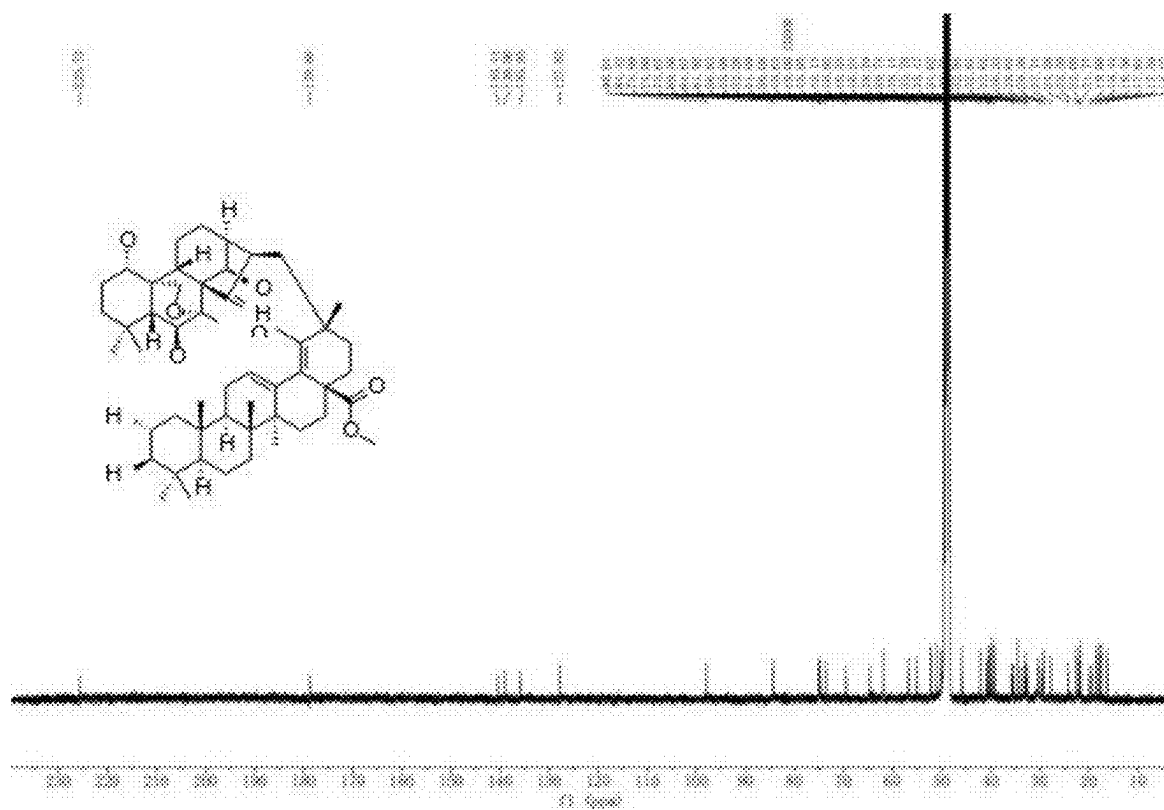


FIG. 4

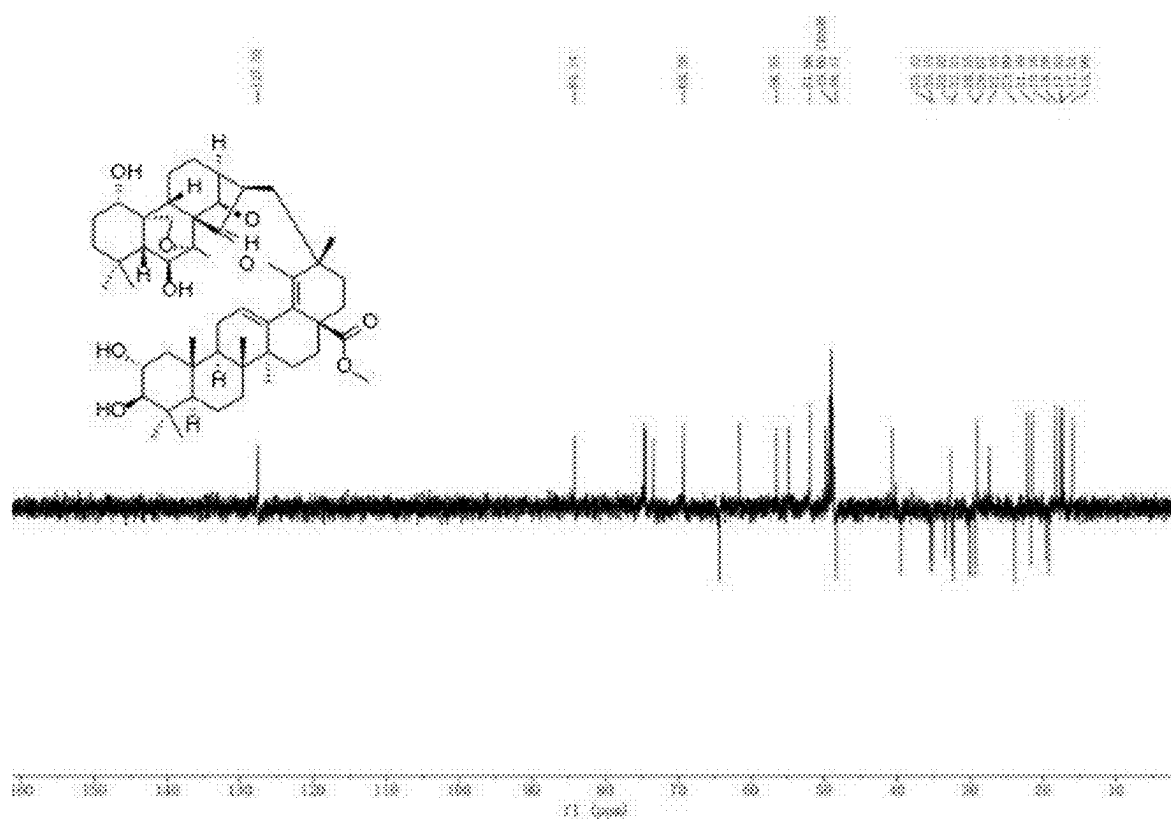


FIG. 5

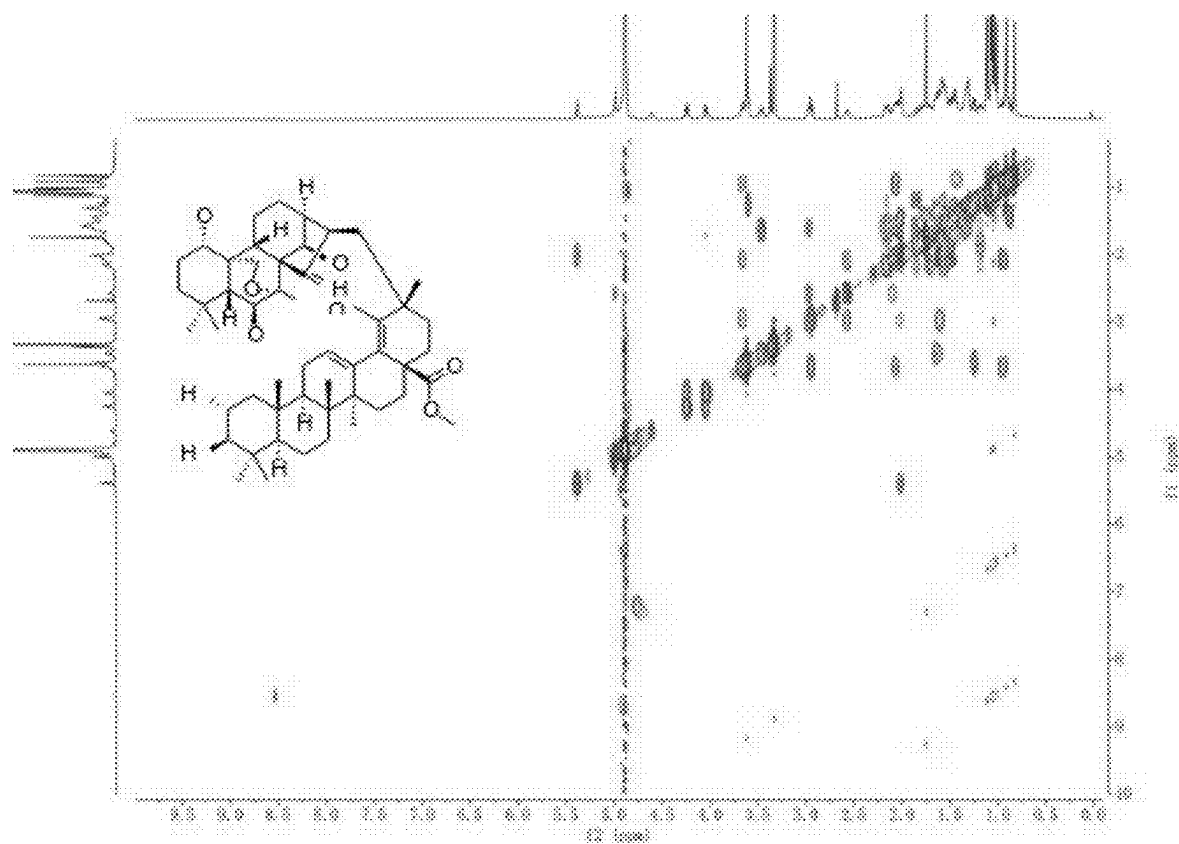


FIG. 6

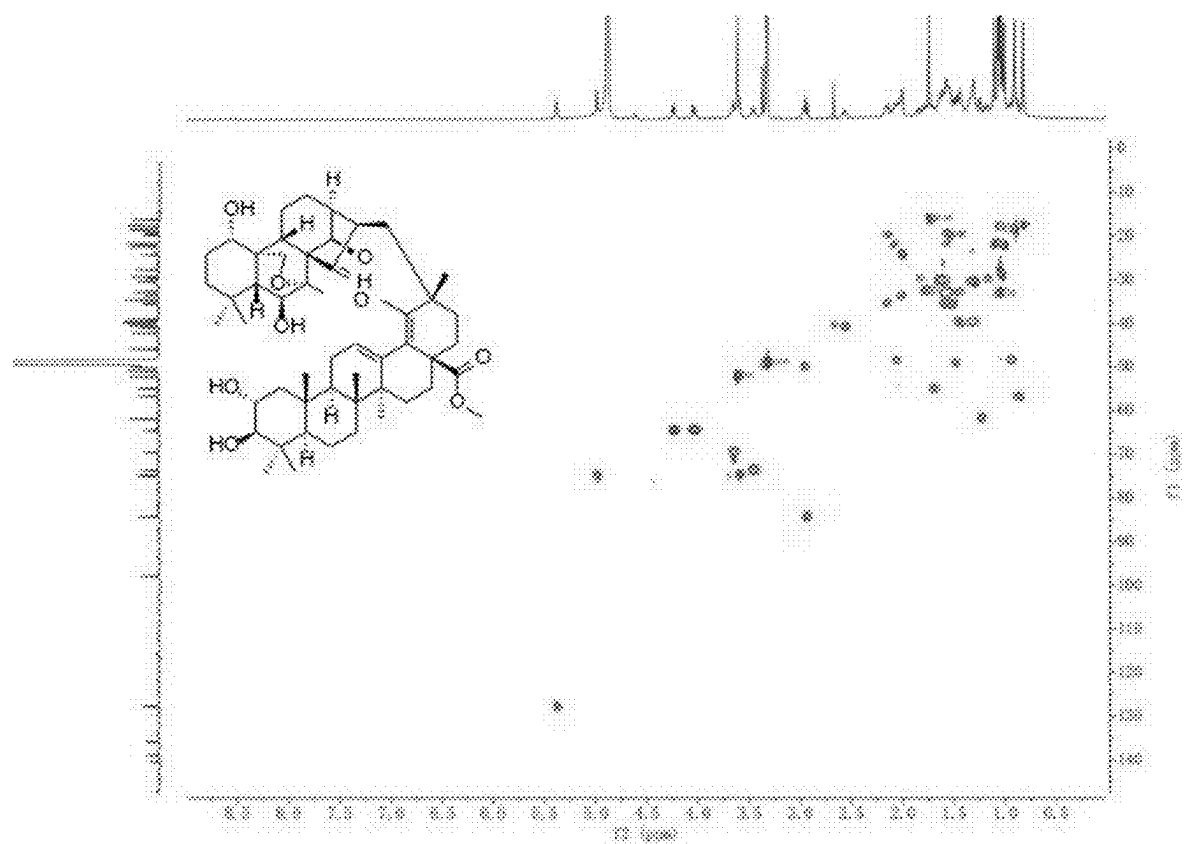


FIG. 7

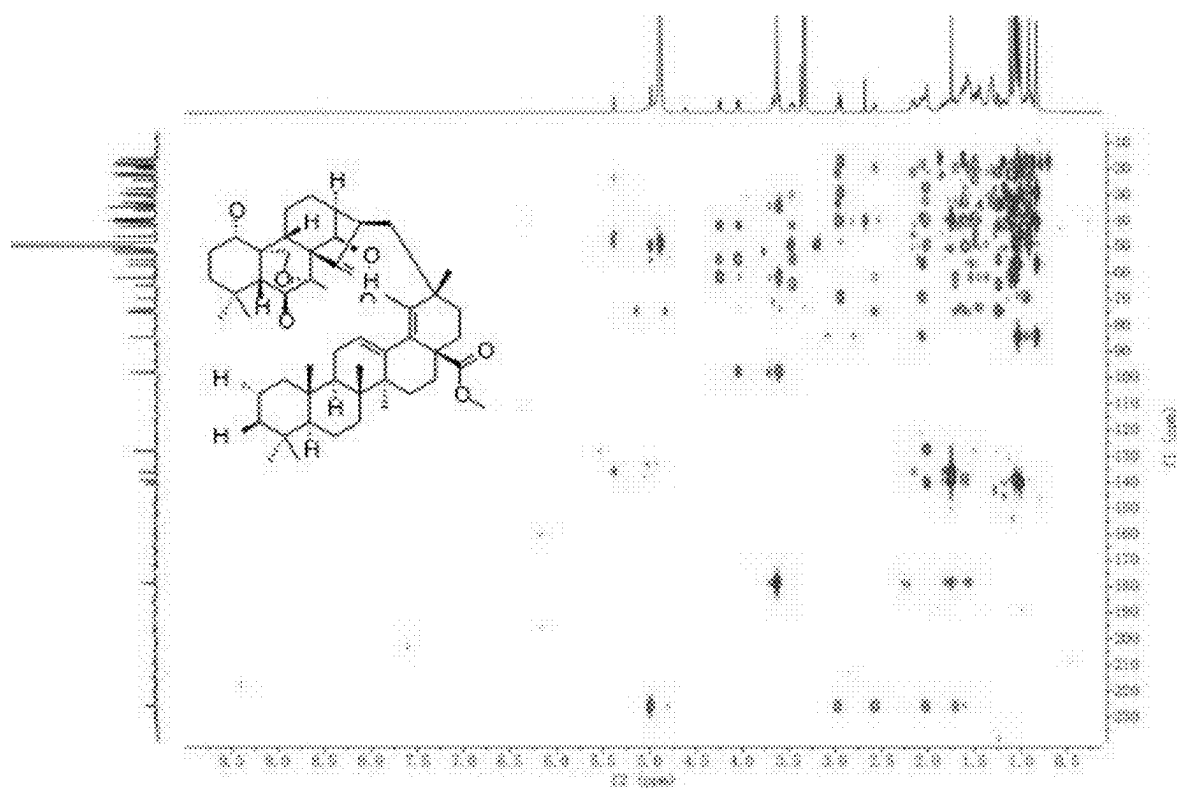


FIG. 8

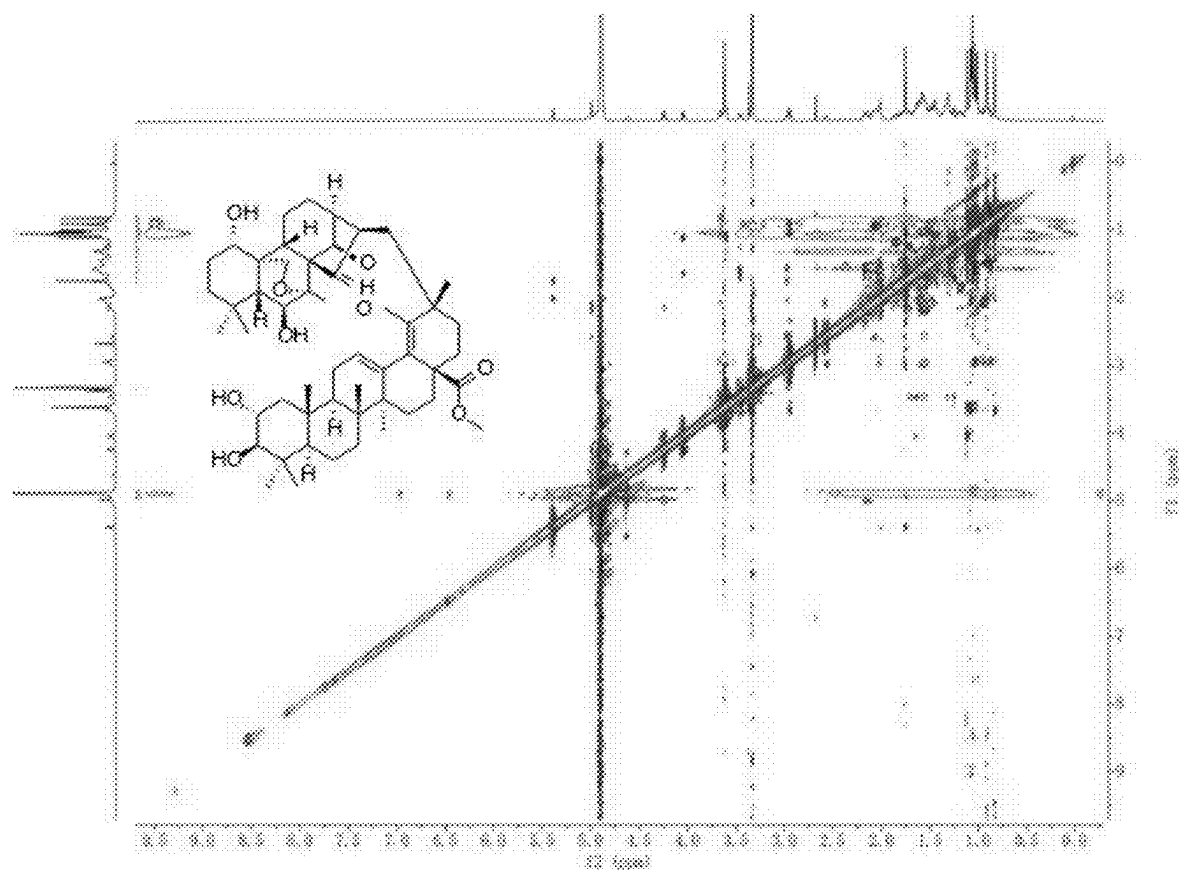


FIG. 9

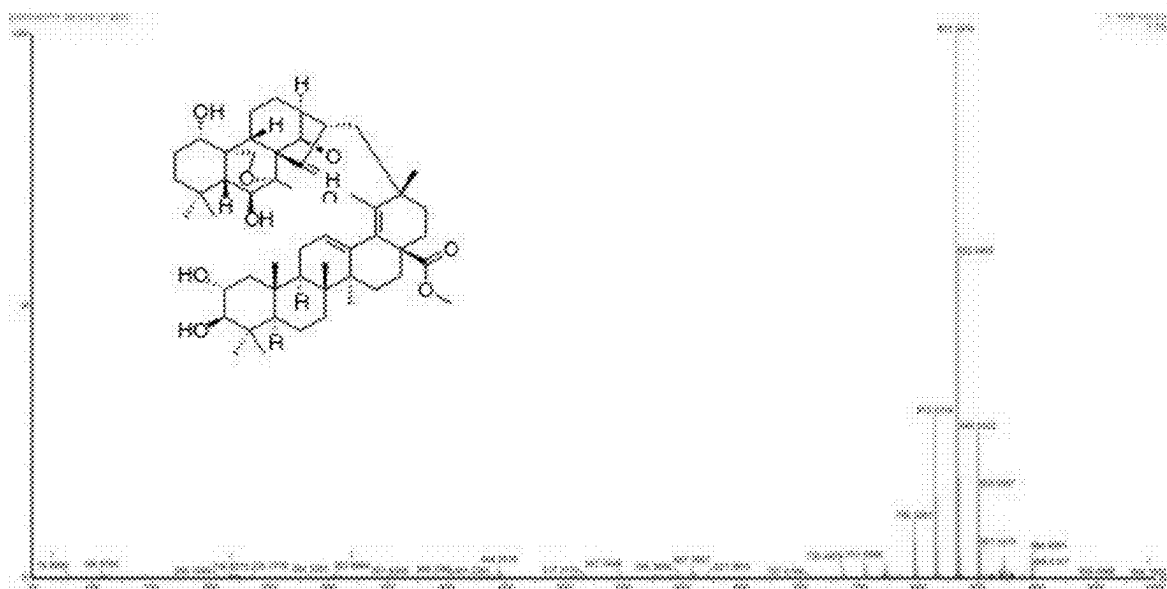


FIG. 10

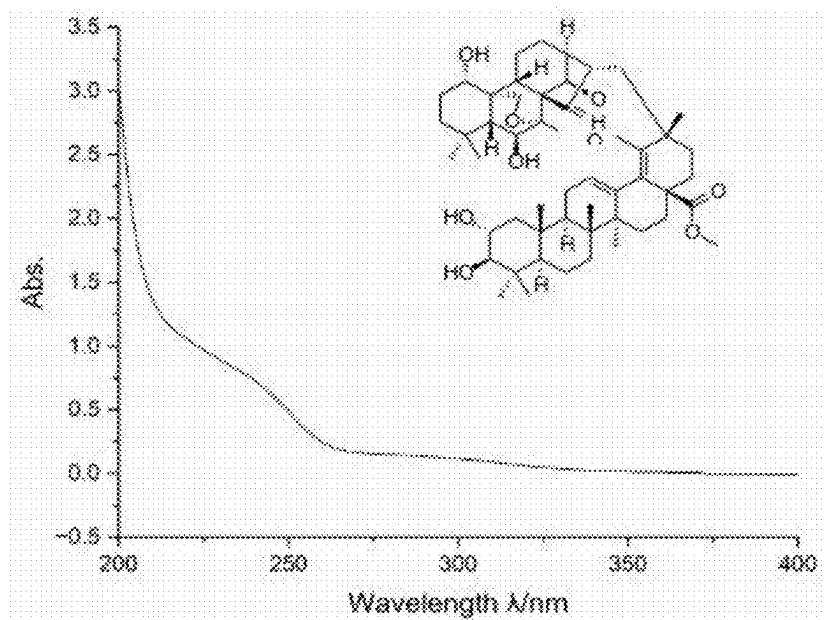


FIG. 11

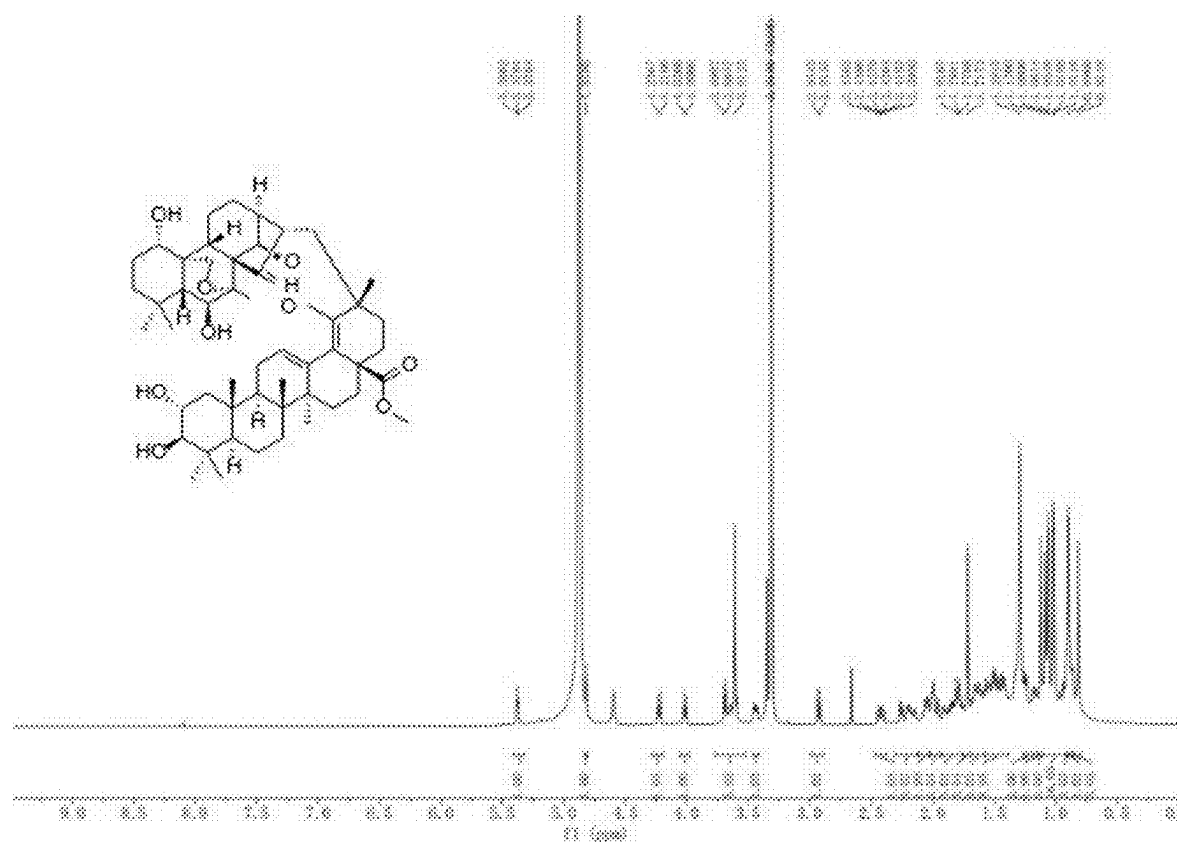
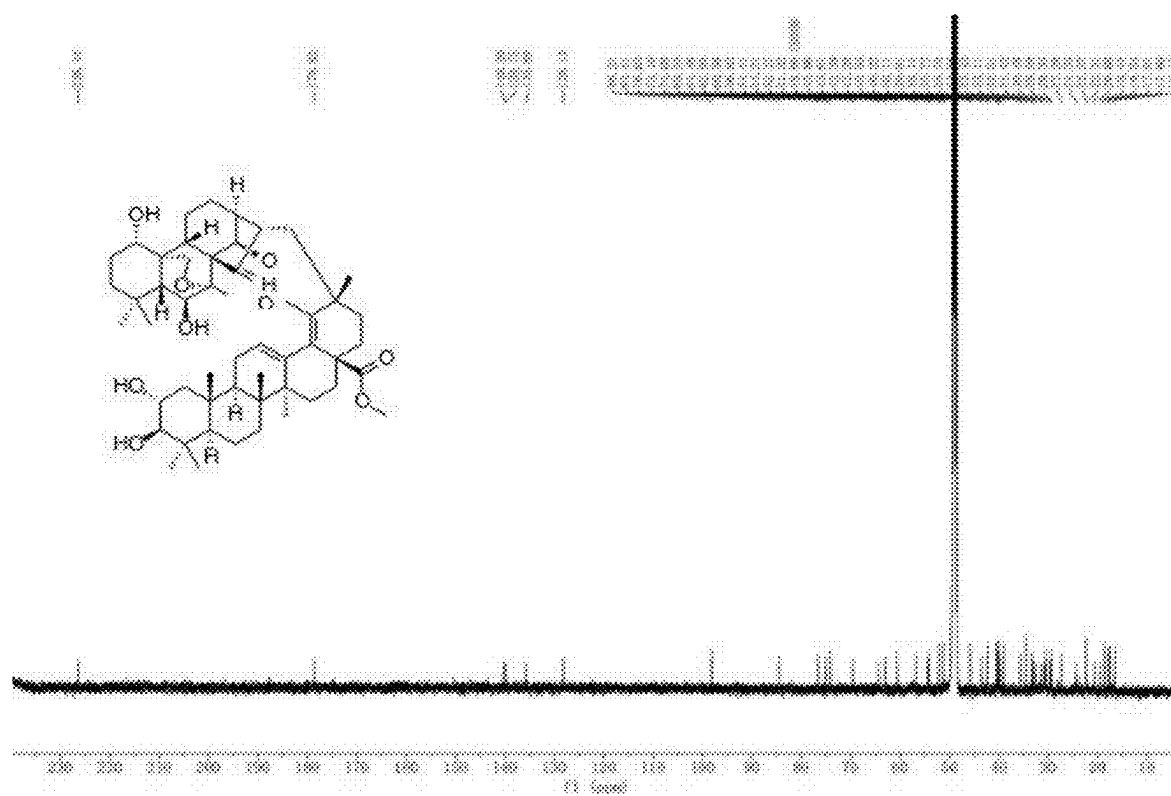


FIG. 12



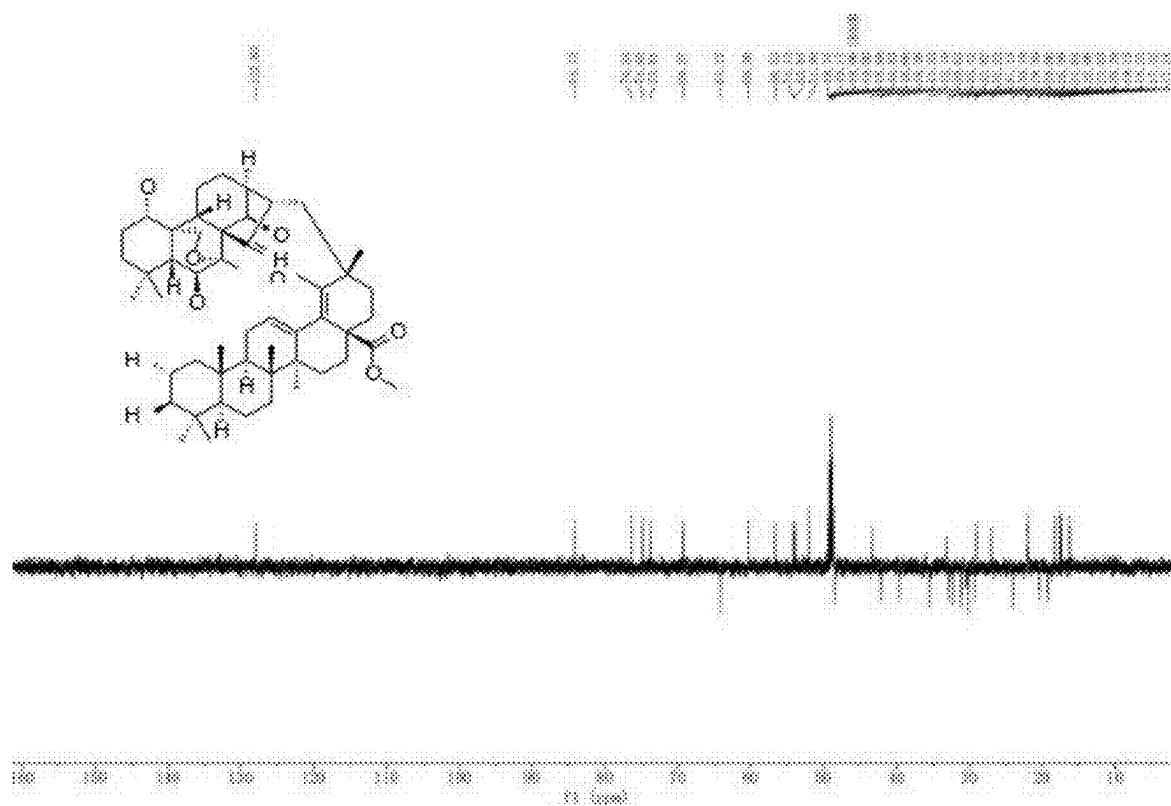


FIG. 14

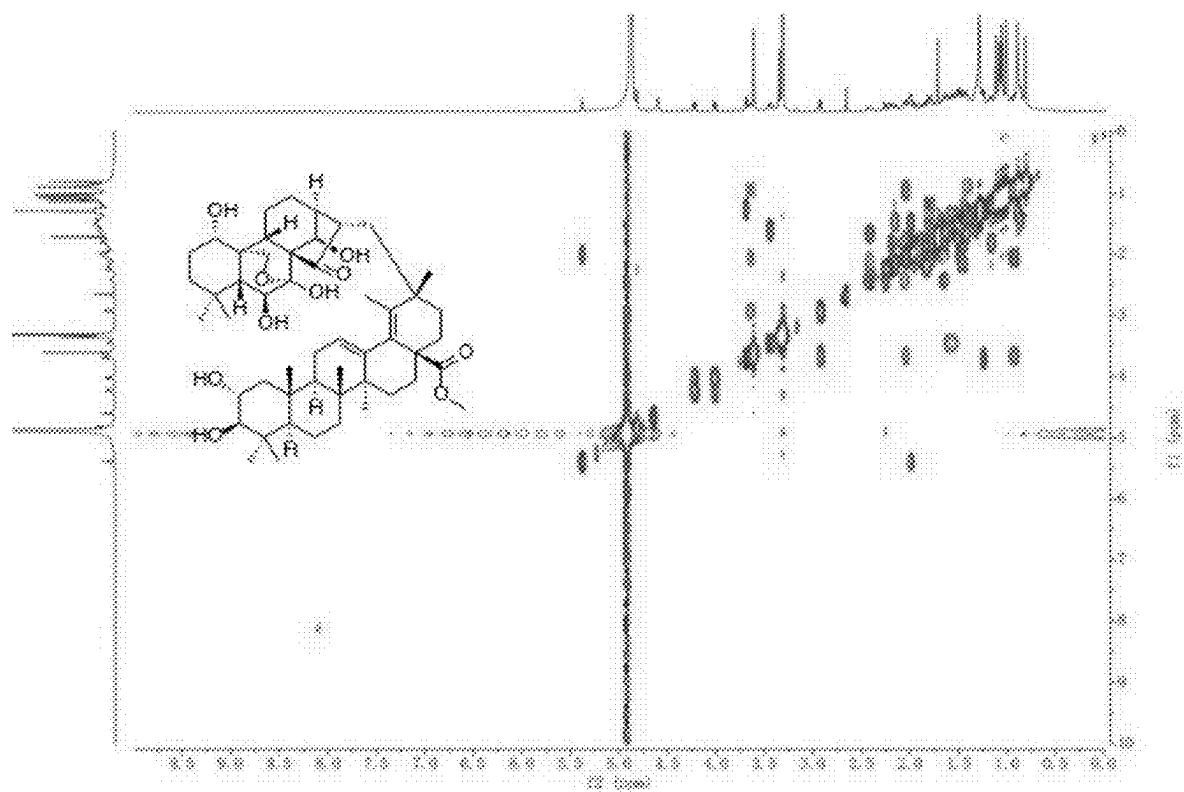


FIG. 15

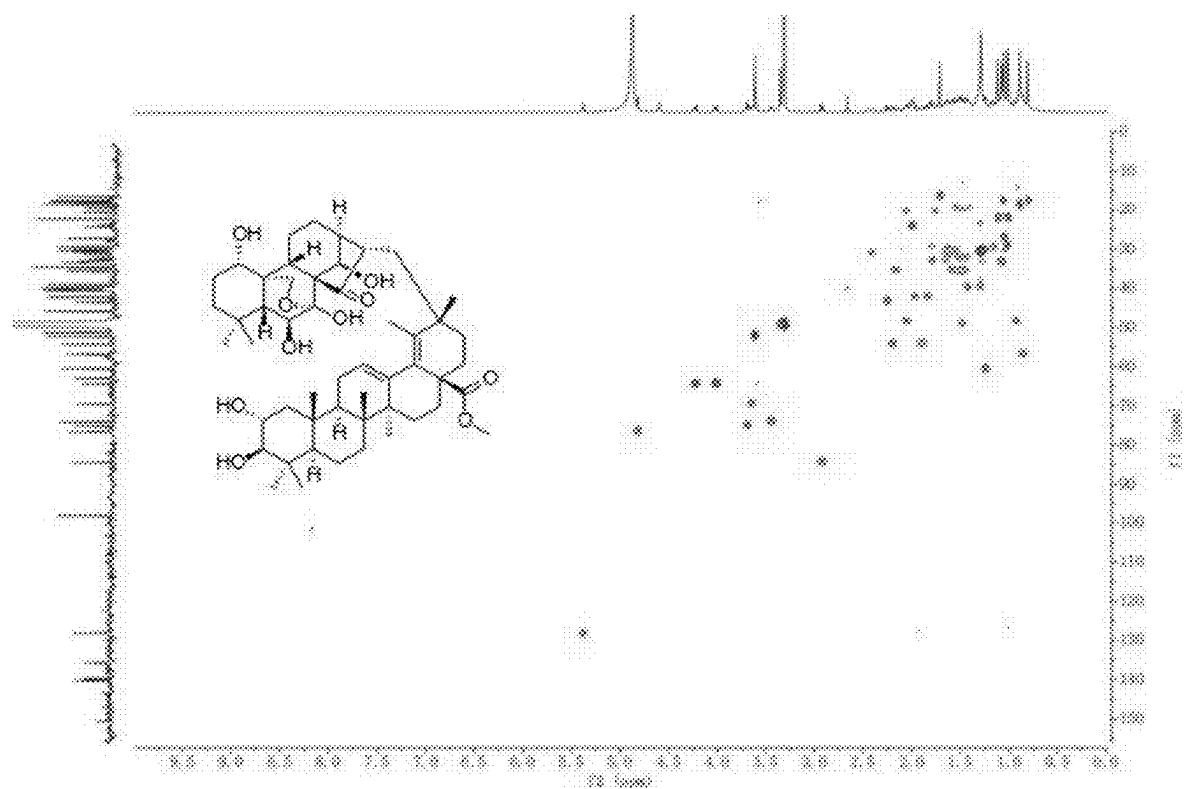


FIG. 16

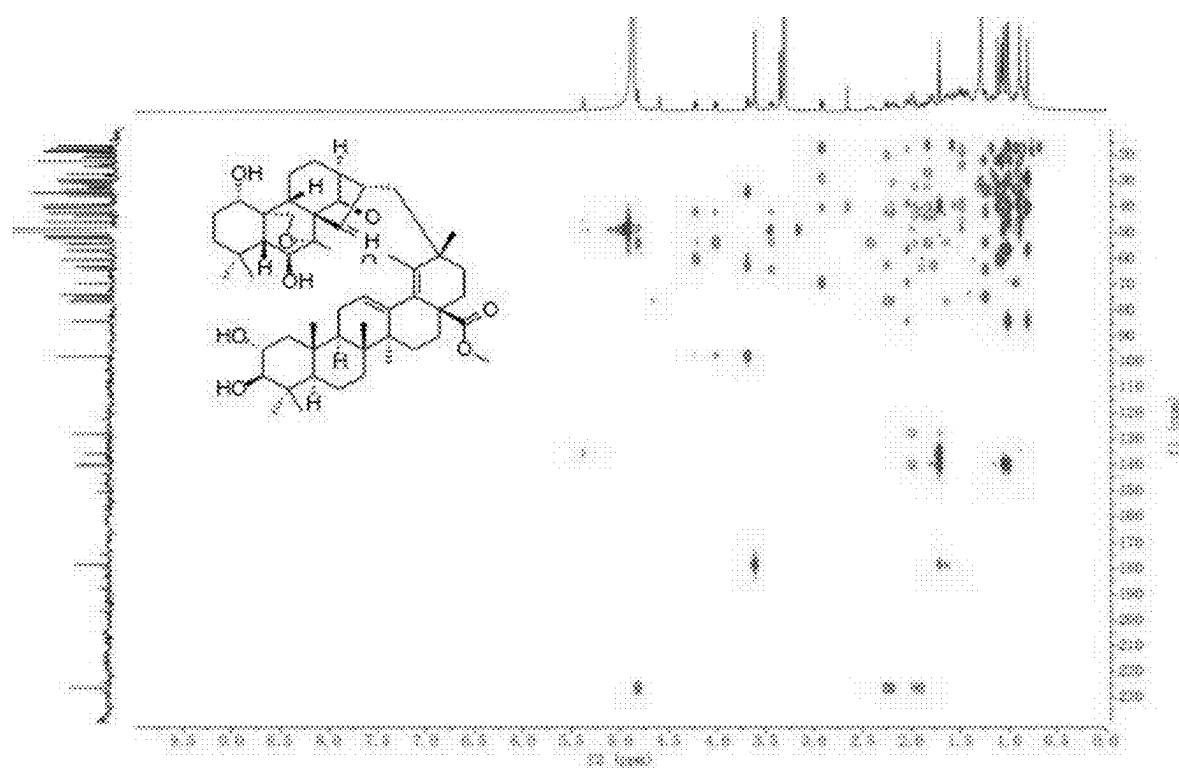


FIG. 17

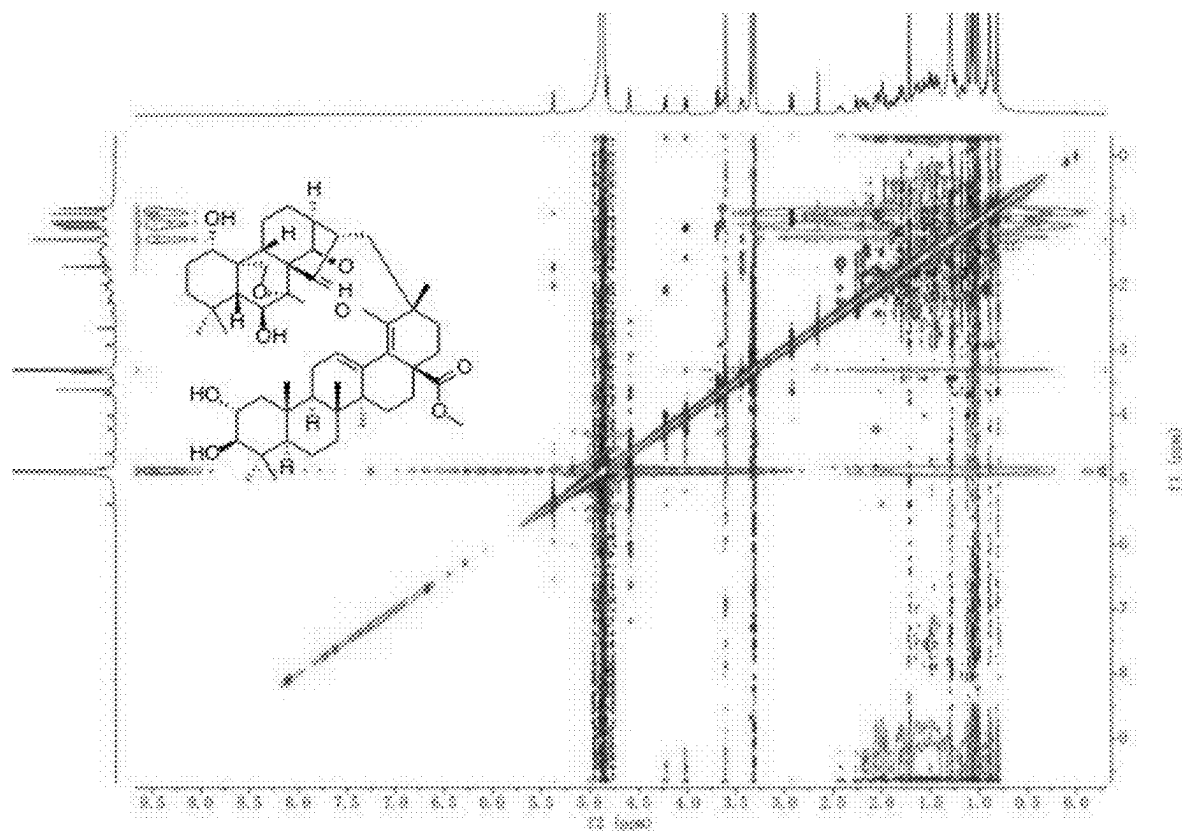


FIG. 18

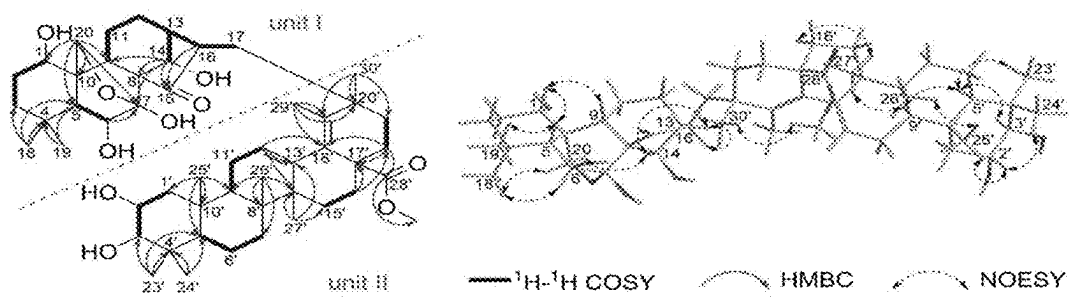


FIG. 19

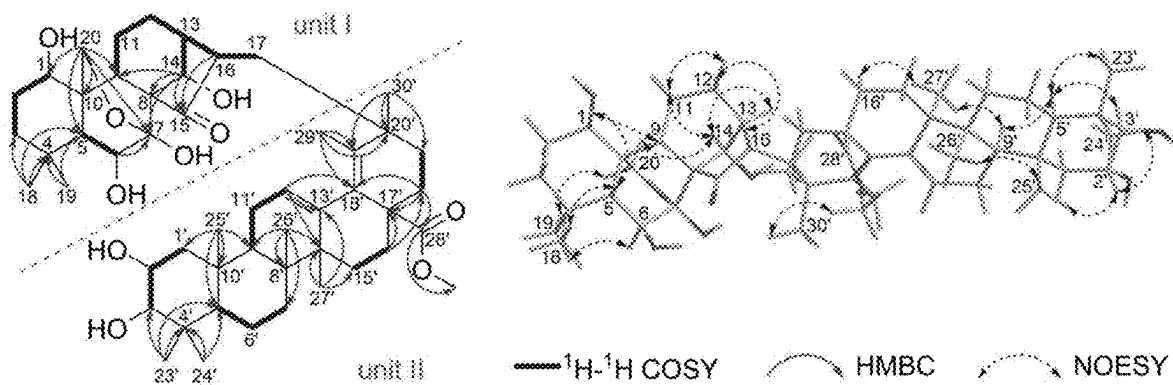


FIG. 20

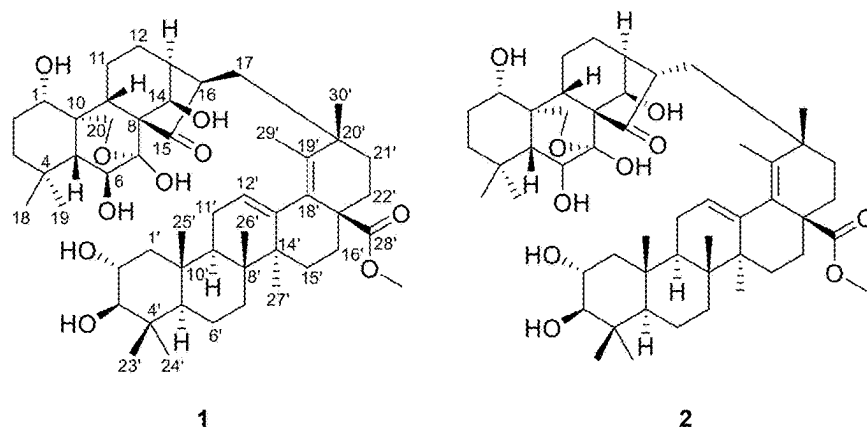


FIG. 21

**COMPOUND WITH NEW SKELETON OF
URSANE TRITERPENOID - ENT-KAURANE
DITERPENOID DIMER, AND PREPARATION
METHOD THEREOF AND APPLICATION
THEREOF**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This application claims priority to Chinese Patent Application Ser. No. CN202510387829X filed on 31 Mar. 2025.

TECHNICAL FIELD

[0002] The present invention belongs to the field of medicine, and particularly to two compounds with a new skeleton of ursane triterpenoid-ent-kaurane diterpenoid dimer, and a preparation method thereof and an application thereof in an anti-tumor effect.

BACKGROUND

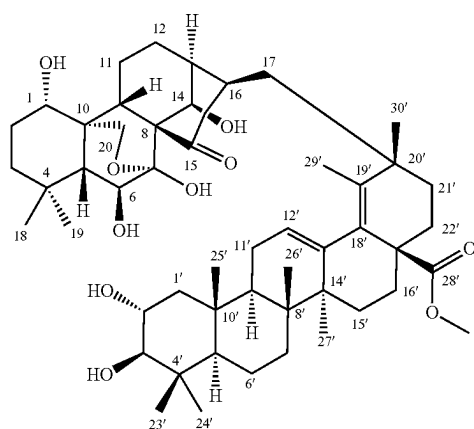
[0003] *I. trichantha* is an endemic plant of *cornus* in Central and West Africa, and is a medicinal plant used by indigenous tribes in Nigeria and neighboring countries. This plant plays an important role in traditional medicine in Nigeria, and its tuber is used as a medicine by local people, which is widely used in the treatment of various diseases, such as poisoning, constipation, emesis, and malaria. The tuber of *I. trichantha* is rich in starch. During the famine years, local people ate the tuber after repeatedly washing it with water. Modern research shows that *I. trichantha* has many physiological activities, such as blood glucose decrease, an anti-convulsion effect, sedation, analgesia, and bacteriostasis. At present, there are few studies on chemical ingredients of *I. trichantha*.

SUMMARY

[0004] The present invention aims to deeply study active ingredients of tuber of *I. trichantha*, and the active ingredients with potential activity are extracted and separated to provide scientific basis for clinical application.

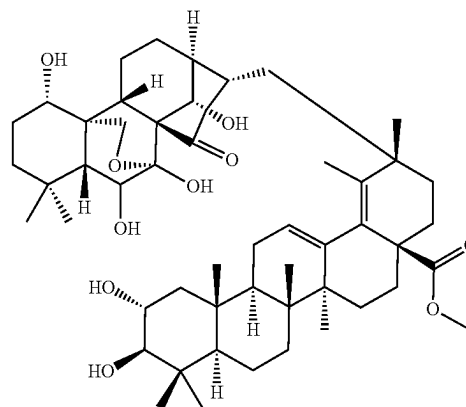
[0005] Technical solution: according to the present invention, two compounds with a new skeleton of ursane triterpenoid-ent-kaurane diterpenoid dimer are separated from the tuber of *I. trichantha*, and the compounds are named Bisicacinal B (1) and Bisicacinal C (2).

[0006] Structural formulae of the compounds 1 and 2 of the present invention are as follows:



1

-continued



2

[0007] A preparation method of the compound with the new skeleton of ursane triterpenoid—ent-kaurane diterpenoid dimer according to the present invention comprises the following steps:

[0008] (1) weighing and crushing dried tuber of *I. trichantha*, adding ethanol aqueous solution, refluxing the mixture for extraction of medicinal material, filtering the mixture and then collecting a filtrate, and concentrating the filtrate under a reduced pressure until no ethanol smell exists to obtain a concentrated solution;

[0009] (2) extracting the concentrated solution obtained in the step (1) with petroleum ether, dichloromethane, ethyl acetate, and n-butanol respectively, and concentrating extracts under a reduced pressure to obtain a petroleum ether fraction, a dichloromethane fraction, an ethyl acetate fraction, an n-butanol fraction, and a raffinate fraction respectively;

[0010] (3) subjecting the dichloromethane fraction obtained in the step (2) to AB-8 macroporous resin atmospheric column chromatography, and eluting the product with ethanol aqueous solution to obtain 3 sub-fractions DCM-1 to DCM-3;

[0011] (4) subjecting the eluate fraction DCM-3 obtained in the step (3) to silica gel atmospheric column chromatography, and eluting the product with dichloromethane-methanol to obtain 7 secondary fractions DCM-3-1 to DCM-3-7; and

[0012] (5) subjecting the fraction DCM-3-5 obtained in the step (4) to medium-pressure preparative-MCI column chromatography, and gradually eluting the product with methanol aqueous solution serving as a mobile phase to obtain 11 fractions DCM-3-5-1 to DCM-3-5-11; and subjecting the fraction DCM-3-5-9 to semi-preparative high-performance liquid chromatography to obtain the compounds 1 and 2.

[0013] As a preferred solution, the preparation method according to the present invention comprises the following steps:

[0014] (1) weighing and crushing dried tuber of *I. trichantha*, adding ethanol at a volume concentration of 80-95%, refluxing the mixture for extraction of medicinal material according to a material-liquid ratio of 1:6-20 for 1-3 times, for 1-3 hours each time, filtering the mixture and then collecting filtrates, combining the

filtrates, and concentrating the combined filtrate under a reduced pressure until no ethanol smell exists to obtain an extractum;

[0015] (2) preparing the extractum obtained in the step (1) into a suspension with a proper amount of water, and then extracting the suspension with petroleum ether, dichloromethane, ethyl acetate, and n-butanol at an equal volume for 3-6 times respectively to obtain a petroleum ether fraction, a dichloromethane fraction, an ethyl acetate fraction, an n-butanol fraction, and a raffinate fraction respectively;

[0016] (3) subjecting the dichloromethane fraction obtained in the step (2) to macroporous resin sample stirring and then to macroporous resin atmospheric column chromatography, and gradiently eluting the product with ethanol aqueous solution serving as a mobile phase at volume ratios of 30:70, 60:40, 90:10, and 100:0 in sequence according to 4-6 column volumes in each gradient elution; and after thin-layer and ultra-high-performance liquid chromatography, concentrating and combining the products by a rotary evaporator to obtain 3 sub-fractions DCM-1 to DCM-3;

[0017] (4) subjecting the eluate fraction DCM-3 obtained in the step (3) to silica gel sample stirring and then to silica gel atmospheric column chromatography, and gradiently eluting the product with dichloromethane-methanol serving as a mobile phase at volume ratios of 100:0, 50:1, 30:1, 20:1, 10:1, 7:1, 5:1, 3:1, 2:1, 1:1, 1:3, 1:7, and 0:100 in sequence according to 3-6 column volumes in each gradient elution; and after thin-layer and ultra-high-performance liquid chromatography, concentrating and combining the products by a rotary evaporator to obtain 7 sub-fractions DCM-3-1 to DCM-3-7;

[0018] (5) subjecting the fraction DCM-3-5 obtained in the step (4) to MCI sample stirring and then to medium-pressure preparative-MCI column chromatography, and gradiently eluting the product with methanol aqueous solution serving as a mobile phase, wherein A is water, and B is methanol; and a gradient elution program is as follows: 0.01-20.00 minutes, 30%-30% B; 20.00-50.00 minutes, 40%-40% B; 50.00-80.00 minutes, 45%-45% B; 80.00-110.00 minutes, 50%-50% B; 110.00-140.00 minutes, 55%-55% B; 140.00-170.00 minutes, 60%-60% B; 170.00-200.00 minutes, 70%-70% B; 200.00-230.00 minutes, 85%-85% B; 230.00-270.00 minutes, 100%-100% B, an elution flow rate of 20 mL/min, and detection wavelengths of 256 nm and 310 nm; and after ultra-high-performance liquid chromatography, concentrating and combining the products to obtain 11 fractions DCM-3-5-1 to DCM-3-5-11; and

[0019] (6) subjecting the eluate fraction DCM-3-5-9 of 200-220 minutes obtained in the step (5) to semi-preparative high-performance liquid chromatography, isocratically eluting the product with pure water A-methanol B serving as a mobile phase, and separating the eluate to obtain the compound 1 and the compound 2.

[0020] As a preferred solution, chromatographic conditions of the semi-preparative high-performance liquid chromatography above are as follows: a chromatographic column model of Heder ODS preparative chromatographic column (10 nm, 5 μ m, 10 $\times$ 250 mm), a column pressure of high-performance liquid chromatography of 9.8 MPa, a

column temperature of 22-26° C., an injection volume of 100 μ L, a mobile phase of MeOH—H₂O (v/v, 80:20), a flow rate of 3 mL/min, and a detection wavelength of 310 nm. The monomer compounds Bisicacinal B (1) and Bisicacinal C (2) are obtained.

Beneficial Effects:

[0021] The present invention deeply studies the chemical ingredients of the tuber of the plant *I. trichantha* in West Africa, which are separated to obtain the two compounds Bisicacinal B (1) and Bisicacinal C (2) with the new skeleton of ursane triterpenoid-ent-kaurane diterpenoid dimer. Pharmaceutical experiments show that the Bisicacinal B (1) and Bisicacinal C (2) prepared by the present invention have significant anti-tumor activity (comprising pancreatic cancer, colon cancer, lung cancer, or liver cancer), and good anti-bacterial activity (comprising *Helicobacter pylori* and *Candida albicans*).

[0022] According to the present invention, the compounds Bisicacinal B (1) and Bisicacinal C (2) obtained by extraction and separation may be prepared into various dosage forms with a pharmaceutically acceptable carrier.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 is a (+)-HR-ESI-MS graph of a compound 1;

[0024] FIG. 2 is a UV graph of the compound 1;

[0025] FIG. 3 is a ¹H NMR graph of the compound 1 (500 MHz, Methanol-d₄);

[0026] FIG. 4 is a ¹³C NMR graph of the compound 1 (125 MHz, Methanol-d₄);

[0027] FIG. 5 is a DEPT 135 graph of the compound 1 (125 MHz, Methanol-d₄);

[0028] FIG. 6 is a ¹H-¹H COSY graph of the compound 1 (500 MHz, Methanol-d₄);

[0029] FIG. 7 is a HSQC graph of the compound 1 (¹H: 500 MHz, ¹³C: 125 MHz, Methanol-d₄);

[0030] FIG. 8 is a HMBC graph of the compound 1 (¹H: 500 MHz, ¹³C: 125 MHz, Methanol-d₄);

[0031] FIG. 9 is a NOSEY graph of the compound 1 (500 MHz, Methanol-d₄);

[0032] FIG. 10 is a (+)-HR-ESI-MS graph of a compound 2;

[0033] FIG. 11 is a UV graph of the compound 2;

[0034] FIG. 12 is a ¹H NMR graph of the compound 2 (500 MHz, Methanol-d₄);

[0035] FIG. 13 is a ¹³C NMR graph of the compound 2 (125 MHz, Methanol-d₄);

[0036] FIG. 14 is a DEPT 135 graph of the compound 2 (125 MHz, Methanol-d₄);

[0037] FIG. 15 is a ¹H-¹H COSY graph of the compound 2 (500 MHz, Methanol-d₄);

[0038] FIG. 16 is a HSQC graph of the compound 2 (¹H: 500 MHz, ¹³C: 125 MHz, Methanol-d₄);

[0039] FIG. 17 is a HMBC graph of the compound 2 (¹H: 500 MHz, ¹³C: 125 MHz, Methanol-d₄);

[0040] FIG. 18 is a NOSEY graph of the compound 2 (500 MHz, Methanol-d₄);

[0041] FIG. 19 a correlation diagram of ¹H-¹H COSY, and key HMBC and NOESY of the compound 1;

[0042] FIG. 20 a correlation diagram of ¹H-¹H COSY, and key HMBC and NOESY of the compound 2; and

[0043] FIG. 21 is structural formulae of the compounds 1 and 2.

DETAILED DESCRIPTION

Embodiment 1

1. Instruments and Materials

1.1 Instruments

Instrument	Manufacturer
ACQUITY UPLC high performance liquid phase	Waters Company of America
CHEETAH®MP medium-pressure preparative chromatography	Tianjin Agela Technologies Co., Ltd.
Nuclear magnetic resonance instrument Bruker AV-500	Bruker Company of Germany
Shimadzu LC-20 AR preparative high performance liquid chromatographic instrument	Shimadzu Company of Japan
R-300 rotary evaporator	BUCHI Limited Company of Switzerland
Intelligent artificial climate box PRX-150B	Ningbo Saifu Experimental Instrument Co., Ltd.
Quitix 125 D-1CN electronic scale	Sartorius Scientific Instruments Co., Ltd.
Milli-Q Advantage system ultrapure water instrument	Millipore Company of America
1300 series A2 clean bench	Thermo Company of America
EnSpire multifunctional microplate reader	PerkinElmer Company of America
S-B50L vertical pressure steam sterilizer	Jiangsu Jintan Medical Instrument Factory
Adjustable pipette	Eppendorf Company of America
Microporous plate thermostatic oscillator	Hangzhou Miu Instruments Co., Ltd.
7500 Real Time PCR System	Applied Biosystems Company of America
IS-RDV1 constant temperature oscillator	Crystal Technology & Industries, Inc. of America
PrimoStar inverted microscope	ZEISS Company of Germany
SIM-F140AY65-PC ice machine	Panasonic Co., Ltd. of Japan
Forma series II water jacket CO ₂ incubator	Thermo Company of America
TDL-80-2B centrifugal machine	Shanghai Anting Co., Ltd.

1.2 Experimental Materials

[0044] Hedera ODS preparative chromatographic column (10 nm, 5 μ m, 10 mm $\times$ 250 mm); Waters ACQUITY UPLC BEH C₁₈ (2.1 mm $\times$ 100 mm, 1.7 μ m) chromatographic column; MCI GEL (CHP20, 75 μ m-150 μ m), and column chromatography silica gel (200-300 meshes); chromatographic acetonitrile, methanol and formic acid purchased from Merck Company of America; and analytical petroleum ether and ethyl acetate purchased from Nanjing

Wanqing Chemical Reagent Co., Ltd.

[0045] A preparation method of a compound was implemented by the following steps.

[0046] (1) Dried tuber of *I. trichantha* was weighed and crushed, and added with ethanol aqueous solution at a volume concentration of 95%, the mixture was refluxed for

extraction of medicinal material according to a material-liquid ratio of 1:10 for 3 times, for 2.5 hours each time, and filtered, then filtrates were collected and combined, and the combined filtrate was concentrated under a reduced pressure until no ethanol smell existed to obtain an extractum.

[0047] (2) The extractum obtained in the step (1) was prepared into a suspension with a proper amount of water, and then the suspension was extracted with petroleum ether, dichloromethane, ethyl acetate, and n-butanol at an equal volume for 5 times respectively to obtain a petroleum ether fraction, a dichloromethane fraction, an ethyl acetate fraction, an n-butanol fraction, and a raffinate fraction respectively.

[0048] (3) The dichloromethane fraction obtained in the step (2) was subjected to sample stirring according to sample: macroporous resin=1:1.5, and then to macroporous resin atmospheric column chromatography, and the product was gradually eluted with ethanol aqueous solution serving as a mobile phase at volume ratios of 30:70, 60:40, 90:10, and 100:0 in sequence according to 5 column volumes in each gradient elution; and after thin-layer and ultra-high-performance liquid chromatography, the products were concentrated and combined by a rotary evaporator to obtain 3 sub-fractions DCM-1 to DCM-3.

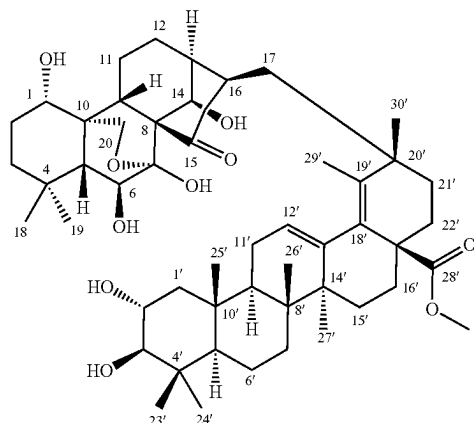
[0049] (4) The eluate fraction DCM-3 obtained in the step (3) was subjected to sample stirring according to sample: silica gel=1:1.5, and then to silica gel atmospheric column chromatography, and the product was gradually eluted with dichloromethane-methanol serving as a mobile phase at volume ratios of 100:0, 50:1, 30:1, 20:1, 10:1, 7:1, 5:1, 3:1, 2:1, 1:1, 1:3, 1:7, and 0:100 in sequence according to 4 column volumes in each gradient elution; and after thin-layer and ultra-high-performance liquid chromatography, the products were concentrated and combined by a rotary evaporator to obtain 7 sub-fractions DCM-3-1 to DCM-3-7.

[0050] (5) The fraction DCM-3-5 obtained in the step (4) was subjected to sample stirring according to sample: MCI=1:1.2, and then to medium-pressure preparative-MCI column chromatography, and the product was gradually eluted with methanol aqueous solution serving as a mobile phase, wherein A was water, and B was methanol; and a gradient elution program was as follows: 0.01-20.00 minutes, 30%-30% B; 20.00-50.00 minutes, 40%-40% B; 50.00-80.00 minutes, 45%-45% B; 80.00-110.00 minutes, 50%-50% B; 110.00-140.00 minutes, 55%-55% B; 140.00-170.00 minutes, 60%-60% B; 170.00-200.00 minutes, 70%-70% B; 200.00-230.00 minutes, 85%-85% B; 230.00-270.00 minutes, 100%-100% B, an elution flow rate of 20 mL/min, and detection wavelengths of 256 nm and 310 nm; and after ultra-high-performance liquid chromatography, the products were concentrated and combined to obtain 11 fractions DCM-3-5-1 to DCM-3-5-11.

[0051] (6) The eluate DCM-3-5-9 of 200-220 minutes obtained in the step (5) was subjected to semi-preparative high-performance liquid chromatography, the product was isocratically eluted with pure water (A)-methanol (B) serving as a mobile phase (MeOH—H₂O, v/v, 80:20), the monomer compound 1 was collected in a time period of 52-57 minutes, and the monomer compound 2 was collected in a time period of 58-61 minutes.

3. Structural Analysis of Compounds

3.1 Structural Identification of Bisicacinal B (1)



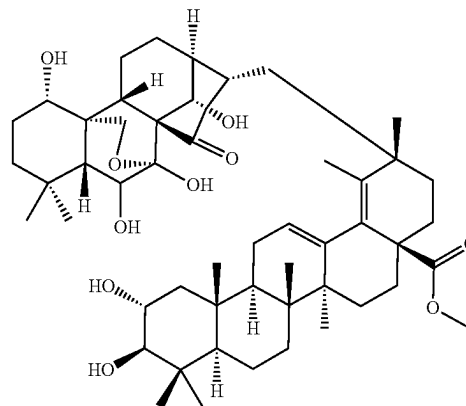
[0052] The compound 1 was white amorphous powder, $[\alpha]_D^{20} +15.0$ (c 0.01, MeOH). In combination with ^{13}C NMR data and a quasi-molecular ion peak m/z 849.5523 ($[\text{M}+\text{H}]^+$, calculated value of $\text{C}_{51}\text{H}_{76}\text{O}_{10}^+$ of 849.5517) given by a high-resolution mass spectrum (HRESIMS), it could be inferred that a molecular formula of the compound was $\text{C}_{51}\text{H}_{76}\text{O}_{10}$, and a degree of unsaturation was 14. A hydrogen spectrum of the compound 1 (Table 1) showed that the compound had 9 methyl proton signals δ_H (1.10, s, CH_3 -18; 1.04, s, CH_3 -19; 1.02, s, CH_3 -23'; 0.81, s, CH_3 -24'; 1.06, s, CH_3 -25'; 0.90, s, CH_3 -26'; 0.99, s, CH_3 -27'; 1.73, s, CH_3 -29'; 1.04, s, CH_3 -30'); and 1 methoxy proton signal δ_H (3.62, s, OCH_3 -28'). A ^{13}C NMR spectrum and a DEPT spectrum showed that the compound had 51 carbon signals, comprising 9 methyl carbons, 14 methylene carbons (comprising 1 oxygen-bond methylene), 12 methine carbons (comprising 5 oxygen-bond methines), 13 non-hydrogen-bond carbons (comprising 1 oxygen-bond carbon), and 2 carbonyl carbons. In combination with a carbon spectrum of the compound 1 (Table 1), a DEPT spectrum and a HSQC spectrum showed that the compound 1 had 51 carbon signals, which could be classified as 9 methyl carbons δ_C 21.8 (CH_3 -18), 32.9 (CH_3 -19), 29.4 (CH_3 -23'), 17.6 (CH_3 -24'), 17.8 (CH_3 -25'), 18.6 (CH_3 -26'), 22.5 (CH_3 -27'), 16.3 (CH_3 -29'), and 27.7 (CH_3 -30'); 1 methoxy carbon δ_C 52.3 (OCH_3 -28'); 14 methylene carbons (comprising 1 oxygen-bond methylene) δ_C 30.3 (C-2), 39.8 (C-3), 19.9 (C-11), 22.0 (C-12), 33.7 (C-17), 64.7 (C-20), 48.7 (C-1'), 19.4 (C-6'), 35.7 (C-7'), 39.1 (C-11'), 29.6 (C-15'), 32.6 (C-16'), 32.8 (C-21'), and 35.5 (C-22'); 12 methine carbons (comprising 5 oxygen-bond methines) δ_C 73.7 (C-1), 61.9 (C-5), 74.8 (C-6), 55.1 (C-9), 40.9 (C-13), 75.0 (C-14), 50.1 (C-16), 69.6 (C-2'), 84.4 (C-3'), 56.8 (C-5'), 49.4 (C-9'), and 127.8 (C-12'); 13 non-hydrogen-bond carbons (comprising 1 oxygen-bond carbon) δ_C 34.6 (C-4), 98.0 (C-7), 49.6 (C-8), 42.1 (C-10), 40.2 (C-4'), 39.7 (C-8'), 39.1 (C-10'), 140.4 (C-13'), 46.0 (C-14'), 51.1 (C-17'), 136.0 (C-18'), 139.1 (C-19'), and 40.5 (C-20'); and 2 carbonyl carbons δ_C 225.7 (C-15) and 178.7 (C-28') respectively.

[0053] NMR data of unit I in the compound 1 were highly similar to those of oridonin, and the difference lied in a

methine (δ_C 50.1, C-16) and a methylene (δ_C 33.7, C-17) of the unit I instead of an exocyclic double bond (δ_C 153.4, C-16; 120.6, C-17); and NMR data of unit II in the compound 1 were highly similar to those of goreishic acid, and the difference lied in that there was a methoxy (δ_H 3.62, δ_C 52.3, OCH_3 -28') in the unit II, and C-20 was a non-hydrogen-bond carbon (δ_C 40.5, C-20') instead of a methine (δ_H 2.28, m; δ_C 24.8, C-20'). In a HMBC spectrum, related signals between CH-16 and C-20' and between CH_2 -17 and C-19'/C-20'/ CH_2 -21'/ CH_3 -30' proved that the units I and II were connected by C-17 and C-20' via a single bond.

[0054] A relative configuration of the compound 1 was determined by analyzing a NOESY spectrum (FIG. 9). Relevant reports have identified H-5 and H-5' as being β -orientated so far, and related signals in the NOESY proved that CH_3 -19/H-58/H-1/H-9 was β -orientated; cross peaks of H-6/ CH_3 -18/H-20a and H-20b/H-14/H-13/H-16 showed that H-6/H-13/H-14/H-16/ CH_3 -18 was α -orientated, and an a correlation between H-5'/ CH_3 -23' and H-3'/H-5'/H-9'/ CH_3 -27'/H-16' showed that H-3'/H-5'/H-9'/ CH_3 -23'/ CH_3 -27' was α -orientated and 28- COOCH_3 was β -orientated; a correlation between 28- COOCH_3 and CH_3 -30' showed that CH_3 -30' was also β -orientated; and it was inferred from a correlation of CH_3 -24'/H-2'/ CH_3 -25'/ CH_3 -26' that H-2', CH_3 -24', CH_3 -25', and CH_3 -26' were 8-orientated. By the measurement and calculation of ECD, it was concluded that an absolute configuration of C-16 was an R configuration, and finally, it was determined that an absolute configuration of the compound 1 was 1S, 5R, 6S, 7S, 8R, 9S, 10S, 13S, 14R, 16R, 2'R, 3'R, 5'R, 8'R, 9'R, 10'R, 14'S, 17'S, and 20'S. Therefore, a complete structure of the compound 1 with the new skeleton was determined, and named Bisicacinal B.

3.2 Structural Identification of Bisicacinal C (2)



[0055] The compound 2 was white amorphous powder, $[\alpha]_D^{20} +15.0$ (c 0.01, MeOH). In combination with ^{13}C NMR data and a quasi-molecular ion peak m/z 849.5522 ($[\text{M}+\text{H}]^+$, calculated value of $\text{C}_{51}\text{H}_{76}\text{O}_{10}^+$ of 849.5517) given by a high-resolution mass spectrum (HRESIMS), it could be inferred that a molecular formula of the compound was $\text{C}_{51}\text{H}_{76}\text{O}_{10}$, and a degree of unsaturation was 14. NMR measurement data of the compound were highly similar to those of the compound 1 (Table 1), and the difference lied in that H-16 (δ_H 2.20) and C-16 (δ_C 54.4) in the compound 2 were quite different from H-16 (δ_H 2.93) and C-16 (δ_C 50.1)

in the compound 1, and it was inferred that the compound 2 was a 16-epimer of the compound 1. In addition, a coupling constant ($J=9.7$ Hz, d) of H-16 in the compound 2 was different from that in the compound 1 ($J=7.2$ Hz t), which also supported the above inference. By the measurement and calculation of ECD, it was determined that C-16 of the compound 2 had an S configuration, and finally, it was determined that an absolute configuration of the compound 2 was 1S, 5R, 6S, 7S, 8R, 9S, 10S, 13S, 14R, 16S, 2'R, 3'R, 5'R, 8'R, 9'R, 10'R, 14'S, 17'S, and 20'S. Therefore, a complete structure of the compound 2 with the new skeleton was determined, and named Bisicacanol C. The compounds 1 and 2 were the first cases of ent-kaurane diterpenoid-ursane triterpenoid heterodimer.

TABLE 1

¹ H and ¹³ C NMR data of compounds 1 and 2 (¹ H, 500 MHz; ¹³ C, 125 MHz)				
position	1		2	
	δ_H , (J in Hz)	δ_C , type	δ_H , (J in Hz)	δ_C , type
1	3.44, dd (10.5, 6.3)	73.7, CH	3.44, dd (11.5, 5.6)	74.1, CH
2 α	1.31, overlapped	30.3, CH ₂	1.57, m	30.5, CH ₂
2 β	1.61, dd (10.4, 4.2)		1.62, m	
3 α	1.31, overlapped	39.8, CH ₂	1.42, m	39.9, CH ₂
3 β	1.44, overlapped		1.30, m	
4		34.6, C		34.6, C
5	1.23, d (6.0)	61.9, CH	1.25, d (7.2)	60.6, CH
6	3.59, d (5.9)	74.8, CH	3.69, d (7.0)	75.1, CH
7		98.0, C		98.2, C
8		49.6, C		48.8, C
9	1.67, dd (12.1, 5.2)	55.1, CH	1.90, dd (13.1, 5.5)	54.2, CH
10		42.1, C		42.4, C
11 α	2.14, dt (12.9, 3.5)	19.9, CH ₂	2.05, dd (11.6, 3.7)	20.7, CH ₂
11 β	1.54, m		1.75, m	
12 α	1.54, m	22.0, CH ₂	2.42, dt (13.7, 8.8)	31.4, CH ₂
12 β	2.06, dd (12.6, 4.7)		1.64, m	
13	2.55, t (8.6)	40.9, CH	2.26, d (9.2)	43.6, CH
14	4.97, d (1.9)	75.0, CH	4.82, s	76.6, CH
15		225.7, C		226.4, C
16	2.93, d (9.2)	50.1, CH	2.20, d (7.2)	54.4, CH
17 α	2.06, m	33.7, CH ₂	1.96, m	42.3, CH ₂
17 β	1.57, m		1.84, d (6.9)	
18	1.10, s	21.8, CH ₃	1.12, s	22.3, CH ₃
19	1.04, s	32.9, CH ₃	1.08, s	33.4, CH ₃
20a	4.23, d (10.1)	64.7, CH ₂	4.23, d (10.0)	64.5, CH ₂
20b	4.03, d (10.3)		4.02, d (10.0)	
1'a	0.94, d (11.9)	48.7, CH ₂	0.95, d (12.3)	48.3, CH ₂
1' β	2.02, dd (12.6, 4.7)		2.05, dd (11.6, 3.7)	
2'	3.64, ddd (12.9, 9.7, 4.7)	69.6, CH	3.65, overlapped	69.6, CH
3'	2.93, d (9.7)	84.4, CH	2.92, d (9.5)	84.4, CH
4'		40.2, C		40.2, C
5'	0.86, d (11.6)	56.8, CH	0.86, overlapped	56.8, CH
6' α	1.54, m	19.4, CH ₂	1.53, m	19.4, CH ₂
6' β	1.44, m		1.42, m	
7' α	1.51, m	35.7, CH ₂	1.57, m	35.7, CH ₂
7' β	1.57, m		1.48, m	
8'		39.7, C		39.1, C
9'	1.48, dd (12.0, 3.5)	49.4, CH	1.48, m	49.2, CH

TABLE 1-continued

¹ H and ¹³ C NMR data of compounds 1 and 2 (¹ H, 500 MHz; ¹³ C, 125 MHz)				
position	1		2	
	δ_H , (J in Hz)	δ_C , type	δ_H , (J in Hz)	δ_C , type
10'		39.1, C		40.5, C
11' α	1.98, d (3.8)	24.3, CH ₂	1.99, dd (8.7, 3.8)	24.3, CH ₂
11' β	2.00, d (3.8)		1.29, overlapped	
12'	5.37, t (3.9)	127.8, CH	5.38, t (3.9)	128.1, CH
13'		140.4, C		140.2, C
14'		46.0, C		45.9, C
15' α	1.81, td (13.3, 3.7)	29.6, CH ₂	1.79, dd (13.9, 4.8)	29.6, CH ₂
15' β	1.14, dt (8.1, 3.5)		1.17, dd (14.1, 4.1)	
16' α	1.70, m	32.6, CH ₂	2.16, dd (13.3, 3.5)	35.7, CH ₂
16' β	1.78, m		1.47, m	
17'		51.1, C		51.3, C
18'		136.0, C		135.7, C
19'		139.1, C		139.8, C
20'		40.5, C		40.1, C
21'a	1.56, m	32.8, CH ₂	1.79, dd (13.9, 4.8)	32.5, CH ₂
21' β	1.73, m		1.64, dd (13.9, 3.4)	
22' α	2.14, dt (12.9, 3.5)	35.5, CH ₂	1.47, m	33.1, CH ₂
22' β	1.57, m		1.52, m	
23' α	1.02, s	29.4, CH ₃	1.01, s	29.4, CH ₃
24' β	0.81, s	17.6, CH ₃	0.81, s	17.6, CH ₃
25'	1.06, s	17.8, CH ₃	1.05, overlapped	17.8, CH ₃
26'	0.90, s	18.6, CH ₃	0.90, s	18.5, CH ₃
27'	0.99, s	22.5, CH ₃	1.00, overlapped	22.4, CH ₃
28'		178.7, C		178.6, C
29'	1.73, s	16.3, CH ₃	1.72, s	16.5, CH ₃
30'	1.04, s	27.7, CH ₃	1.05, overlapped	27.3, CH ₃
28'-OCH ₃	3.62, s	52.3, CH ₃	3.61, s	52.3, CH ₃

Embodiment 2

[0056] An anti-tumor cell activity test research of the present invention was carried out by the following steps.

1. Culture of Tumor Cells

[0057] A pancreatic cancer cell line MIA PaCa-2, a colon cancer cell line HT-29, a lung cancer cell line A549, and a liver cancer cell line HepG2 (cell bank of Chinese Academy of Sciences) were cultured in a DMEM culture solution containing 10% fetal bovine serum (Gibco Company) at 37° C. under 5% CO₂. HT-29, A549 and HepG2 cell culture media were supplemented with 10% FBS and 1% PSN, and a MIA PaCa-2 cell culture medium was supplemented with 10% FBS, 2.5% HS and 1% PSN.

2. Preparation of Experimental Drugs

[0058] Proper amounts of the compounds 1 and 2 prepared in the above Embodiment 1 were weighed and dissolved in DMSO, so that mother solutions had a final concentration of 40 mM, and were stored in a refrigerator at 4° C. Before the experiment, the mother solutions were diluted with a DMEM culture medium, so as to make drugs have a concentration of 20 μ M and ensure that the DMSO had a

final concentration lower than 0.1%. Different volumes of DMEM culture media were added to dilute the compounds into different concentrations. Meanwhile, a DMEM culture medium containing 0.1% DMSO was used as a negative control.

Toxicities of Drugs to Tumor Cell Lines

[0059] Tumor cells were suspended in a culture medium, MIA PaCa-2, HT-29 and HepG2 cells were inoculated into a 96-well plate at a cell density of 8000, A549 cells were inoculated into a 96-well plate at a cell density of 3000 (100 μ L/well), and the cells were cultured at 37° C. under 5% CO₂ for 24 hours. In a logarithmic growth phase of the tumor cell lines, different concentrations of compounds (0, 0.004, 0.04, 0.4, 4, and 40 μ M) were added to culture the cells at 37° C. under 5% CO₂ for 72 hours.

Detection of Cell Viability by CCK-8 Method

[0060] After 72 hours of interaction between the drugs and the tumor cells, 10 μ L of CCK-8 solution was added into each well to incubate the cells in a sterile incubator for 1 hour, and then the solution was taken out to determine an OD value by a microplate reader at a wavelength of 450 nm. IC₅₀ (median inhibitory concentration) values of the compounds were calculated by GraphPad Prism 8 software, and 5-FU and Carboplatin were used as positive controls.

[0061] Experimental results referred to Table 2.

TABLE 2

Cytotoxic activities of compounds 1 and 2 (IC ₅₀ : μ M)				
no.	IC ₅₀ $\pm$ SD (μ M) ^a			
	MIA PaCa-2	HT-29	A549	HepG2
Compound 1	13.07 $\pm$ 0.18	14.95 $\pm$ 0.14	4.20 $\pm$ 0.24	14.84 $\pm$ 0.12
Compound 2	>40	33.35 $\pm$ 0.06	23.10 $\pm$ 0.12	>40
5-Fluorouracil	/	12.87 $\pm$ 0.04	/	4.82 $\pm$ 0.07
Carboplatin	/	/	>40	/

[0062] Experimental conclusion: the evaluation of cytotoxic activities of the compounds 1 and 2 showed that the compound 1 had a strong activity on the pancreatic cancer cell line MIA PaCa-2, the colon cancer cell line HT-29, the lung cancer cell line A549, and the liver cancer cell line HepG2, which was superior to those of the compound 2 and the positive control drugs 5-FU and Carboplatin. The compound 2 also showed a strong inhibitory activity on the colon cancer cell line HT-29 and the lung cancer cell line A549, which was superior to those of the positive drugs 5-FU and Carboplatin, so that the compound had the potential to develop a new anti-lung cancer drug.

Embodiment 3

[0063] An antibacterial activity test research of the present invention was carried out by the following steps.

1. Experimental Materials and Reagents

[0064] *Helicobacter pylori* (G27 and HP129) and *Candida albicans* (SC5314 and C5) both came from Professor Hongkai Bi's laboratory of Nanjing Medical University. Main culture mediums and main reagents: Colombian cul-

ture medium, LB culture medium, selective antibiotics (metronidazole and amphotericin B), serum, and the like.

2. Preparation of Culture Media

[0065] Colombian liquid culture medium: 29.0 g of Colombian liquid culture medium was accurately weighed, heated, and dissolved in 1000 mL of double distilled water, and the culture medium was subjected to autoclaved sterilization at 121° C. for 15 minutes after being completely dissolved for later use.

[0066] Colombian blood agar solid culture medium: 39.0 g of Colombian blood agar solid culture medium was accurately weighed, heated, and dissolved in 1000 mL of double distilled water, and the culture medium was subjected to autoclaved sterilization at 121° C. for 15 minutes after being completely dissolved, naturally cooled to about 50° C. after the autoclaved sterilization, quickly added with 5% sterile defibrinated sheep blood, mixed evenly, and then poured into a sterile dish while the mixture was hot.

[0067] LB culture medium: 15 g/L glucose, 10 g/L yeast extract, 5 g/L peptone, and 10 g/L sodium chloride.

3. Resuscitation and Culture of Tested Strains

[0068] Standard strains of *Helicobacter pylori* were taken out of a refrigerator at -80° C. and placed at room temperature, 200 μ L of standard strains were accurately sucked, transferred to a solid culture medium, and spread with an L-shaped glass rod, a culture dish loaded with the strains was put into a culture bag, then an AnaeroPack was put into the culture bag, and the culture bag was quickly sealed and then put into a constant-temperature incubator at 37° C. to culture the strains for 72 hours. After the end of culture, the *Helicobacter pylori* was identified first, then a bacterial lawn on the solid culture medium was scraped off, and the bacterial lawn was transferred into 50 mL of liquid culture medium to serve as an original bacterial solution.

4. Determination of MIC

[0069] (1) Sample solutions of the compounds 1 and 2 prepared in Embodiment 1 were configured to have a concentration of 2 mg/mL.

[0070] (2) In preparation by a MIC plate, a first well was added with 173.6 μ L of culture medium and then added with 6.4 μ L of antibacterial drug, and dilution was carried out by multiple proportions to a 7th well; and an 8th well was not added with the drug, but was reserved with 90 μ L of culture medium to serve as a control with bacteria but no drug.

[0071] (3) In preparation of bacterial solution, the *Helicobacter pylori* growing at a logarithmic phase on a solid plate was made into a bacterial suspension with a BHI culture medium, an OD600 value of concentration of the bacterial suspension was adjusted to be 0.3 (1 $\times$ 10⁸ CFU/mL), and then the bacterial suspension was diluted 10 times to reach 1 $\times$ 10⁷ CFU/mL for later use.

[0072] (4) In inoculation, 10 μ L of bacterial solution was added into the 1st to 8th wells (a concentration of the bacterial solution in each well was about 1.0 $\times$ 10⁶ CFU/mL). Results were judged after 72 hours of culture. Drug concentrations from the 1st to 7th wells were 64 μ g/mL, 32 μ g/mL, 16 μ g/mL, 8 μ g/mL, 4 μ g/mL, 2 μ g/mL, and 1 μ g/mL respectively.

[0073] (5) In result judgment, a lowest drug concentration capable of completely inhibiting the growth of bacteria in

the wells was MIC. The test was only meaningful when the bacteria in the 8th well of the positive control well (without antibiotics) grew obviously. When there was a single drift in a microdilution method, a highest drug concentration capable of inhibiting the growth of bacteria should be recorded. If there were many drifts, the results should not be reported, and the test should be repeated. Each drug was tested repeatedly for 3 times.

5. Experimental Results Referred to Tables 3 and 4.

TABLE 3

Inhibitory effects of compounds 1 and 2 on different strains (MIC: $\mu\text{g/ml}$)			
Strain	1	2	MTZ
<i>Helicobacter pylori</i> H. pylori G27	16	16	2
<i>Helicobacter pylori</i> H. pylori 129	32	32	8 (R)

MTZ, metronidazole;
R, drug-resistant strain

TABLE 4

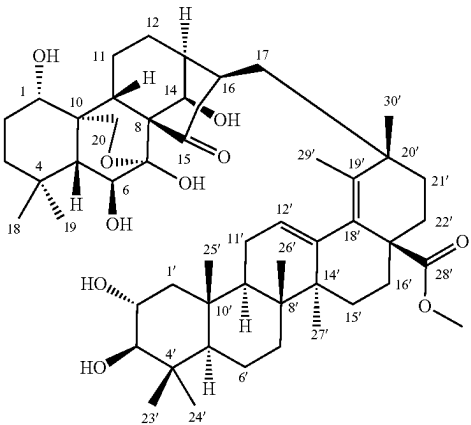
Synergistic inhibitory effect of compound 1 and amphotericin B (AMB) on different strains (MIC: $\mu\text{g/ml}$)							
Strain	Compound 1 ($\mu\text{g/ml}$)		AMB ($\mu\text{g/ml}$)		MIC ($\mu\text{g/ml}$) + FICI index		Mode of action
	Separate use	Combined use	Separate use	Combined use	0.5 $\mu\text{g/ml}$ AMB	after combination	
SC5314	>32	1	2	≤ 0.25	1	≤ 0.078	Synergistic effect
C5	>32	1	2	≤ 0.25	1	≤ 0.078	Synergistic effect

[0074] The above experimental results show that the compounds 1 and 2 have a good antibacterial activity on both sensitive *Helicobacter pylori* and drug-resistant *Helicobacter pylori*, and the MIC is 16-32 $\mu\text{g/mL}$, which indicates that the compounds 1 and 2 have an inhibitory effect on *Helicobacter pylori*. Compound 1 has no significant inhibitory effect on the growth of two kinds of *Candida albicans* when used separately, but when the compound is used in combination with AMB, the MIC of the compound can be greatly reduced, and the MIC of AMB can be reduced below 0.25 $\mu\text{g/mL}$ at the same time, so that the compound can be used to prepare an anti-*Candida albicans* drug or serve as a lead compound for the development of anti-*Candida albicans* drug.

What is claimed is:

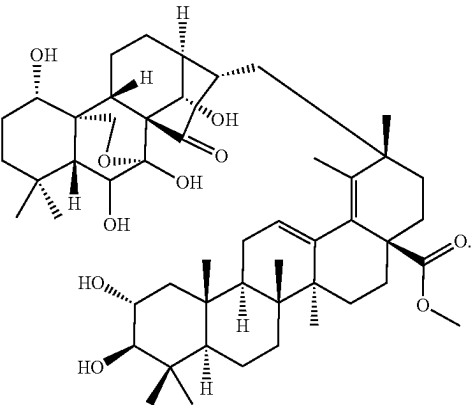
1. A compound with a new skeleton of ursane triterpenoid—ent-kaurane diterpenoid dimer, comprising com-

pounds 1 and 2 as shown by the following structural formulae:



1

-continued



2

2. The compound according to claim 1, wherein the compound and a pharmaceutically acceptable carrier are prepared into a pharmaceutical drug.

3. The compound according to claim 1, wherein the pharmaceutical drug is mixed with an amphotericin B as an anti-*Candida albicans* drug.

4. A preparation method of the compound according to claim 1, comprising the following steps:

- (1) weighing and crushing dried tuber of *I. trichantha*, adding ethanol aqueous solution, refluxing the mixture for extraction of medicinal material, filtering the mixture and then collecting a filtrate, and concentrating the filtrate under a reduced pressure until no ethanol smell exists to obtain a concentrated solution;
- (2) extracting the concentrated solution obtained in the step (1) with petroleum ether, dichloromethane, ethyl acetate, and n-butanol respectively, and concentrating the extracts under a reduced pressure to obtain a petroleum ether fraction, a dichloromethane fraction, an ethyl acetate fraction, an n-butanol fraction, and a raffinate fraction respectively;
- (3) subjecting the dichloromethane fraction obtained in the step (2) to AB-8 macroporous resin atmospheric column chromatography, and eluting the product with ethanol aqueous solution to obtain 3 sub-fractions DCM-1 to DCM-3;
- (4) subjecting the eluate fraction DCM-3 obtained in the step (3) to silica gel atmospheric column chromatography, and eluting the product with dichloromethane-methanol to obtain 7 secondary fractions DCM-3-1 to DCM-3-7; and
- (5) subjecting the fraction DCM-3-5 obtained in the step (4) to medium-pressure preparative-MCI column chromatography, and gradiently eluting the product with methanol aqueous solution serving as a mobile phase to obtain 11 fractions DCM-3-5-1 to DCM-3-5-11; and subjecting the fraction DCM-3-5-9 to semi-preparative high-performance liquid chromatography to obtain the compounds 1 and 2.

5. The preparation method of the compound according to claim 2, comprising the following steps:

- (1) weighing and crushing dried tuber of *I. trichantha*, adding ethanol at a volume concentration of 80-95%, refluxing the mixture for extraction of medicinal material according to a material-liquid ratio of 1:6-20 for 1-3 times, for 1-3 hours each time, filtering the mixture and then collecting filtrates, combining the filtrates, and concentrating the combined filtrate under a reduced pressure until no ethanol smell exists to obtain an extractum;
- (2) preparing the extractum obtained in the step (1) into a suspension with a proper amount of water, and then extracting the suspension with petroleum ether, dichloromethane, ethyl acetate, and n-butanol at an equal volume for 3-6 times respectively to obtain a petroleum ether fraction, a dichloromethane fraction, an ethyl acetate fraction, an n-butanol fraction, and a raffinate fraction respectively;
- (3) subjecting the dichloromethane fraction obtained in the step (2) to macroporous resin sample stirring and then to macroporous resin atmospheric column chromatography, and gradiently eluting the product with ethanol aqueous solution serving as a mobile phase at volume ratios of 30:70, 60:40, 90:10, and 100:0 in sequence according to 4-6 column volumes in each gradient elution; and after thin-layer and ultra-high-

performance liquid chromatography, concentrating and combining the products by a rotary evaporator to obtain 3 sub-fractions DCM-1 to DCM-3;

- (4) subjecting the eluate fraction DCM-3 obtained in the step (3) to silica gel sample stirring and then to silica gel atmospheric column chromatography, and gradiently eluting the product with dichloromethane-methanol serving as a mobile phase at volume ratios of 100:0, 50:1, 30:1, 20:1, 10:1, 7:1, 5:1, 3:1, 2:1, 1:1, 1:3, 1:7, and 0:100 in sequence according to 3-6 column volumes in each gradient elution; and after thin-layer and ultra-high-performance liquid chromatography, concentrating and combining the products by a rotary evaporator to obtain 7 sub-fractions DCM-3-1 to DCM-3-7;
- (5) subjecting the fraction DCM-3-5 obtained in the step (4) to MCI sample stirring and then to medium-pressure preparative-MCI column chromatography, and gradiently eluting the product with methanol aqueous solution serving as a mobile phase, wherein A is water, and B is methanol; and a gradient elution program is as follows: 0.01-20.00 minutes, 30%-30% B; 20.00-50.00 minutes, 40%-40% B; 50.00-80.00 minutes, 45%-45% B; 80.00-110.00 minutes, 50%-50% B; 110.00-140.00 minutes, 55%-55% B; 140.00-170.00 minutes, 60%-60% B; 170.00-200.00 minutes, 70%-70% B; 200.00-230.00 minutes, 85%-85% B; 230.00-270.00 minutes, 100%-100% B, an elution flow rate of 20 mL/min, and detection wavelengths of 256 nm and 310 nm; and after ultra-high-performance liquid chromatography, concentrating and combining the products to obtain 11 fractions DCM-3-5-1 to DCM-3-5-11; and
- (6) subjecting the eluate fraction DCM-3-5-9 of 200-220 minutes obtained in the step (5) to semi-preparative high-performance liquid chromatography, isocratically eluting the product with pure water A-methanol B serving as a mobile phase, and separating the eluate to obtain the compound 1 and the compound 2.

6. The preparation method of the compound according to claim 3, wherein chromatographic conditions of semi-preparative high-performance liquid chromatography are as follows: a chromatographic column model of Heder ODS preparative chromatographic column, specifications of 10 nm, 5 μ m, and 10 $\times$ 250 mm, a column pressure of high-performance liquid chromatography of 9.8 MPa, a column temperature of 22-26 $^{\circ}$ C., an injection volume of 100 μ L, a mobile phase of MeOH and H₂O at a volume ratio of 80:20, a flow rate of 3 mL/min, and a detection wavelength of 310 nm.

7. A method for treating a disease comprising a step of administering the compound according to claim 1 to a subject in need, wherein the disease is a cancer or an infective disease.

8. The method according claim 7, wherein the cancer is selected from the group consisting of a pancreatic cancer, a colon cancer, a lung cancer and a liver cancer.

9. The method according claim 7, wherein the infective disease is caused from a bacterial or a fungal.

10. The method according claim 9, wherein the bacteria is *Helicobacter pylori*.

11. The method according claim 9, wherein the fungal is *Candida albicans*.

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